

ALLEN CHRISTOPHER

Production Engineer | Manufacturing Systems | Digital Twin & Industry 5.0
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PROFESSIONAL SUMMARY

Production Engineering graduate student at Chalmers with experience in manufacturing systems, process optimisation, and digital tools including SAP, Power BI, and Siemens Plant Simulation. Recent experience at Nouryon in process automation and data-driven operations. Currently developing a human-centric digital twin integrating real-time data and simulation to enhance operator decision-making. Experience in developing real-time digital twin systems integrating live data streams, human motion tracking, and safety monitoring. Motivated to apply production engineering and digitalisation to improve manufacturing performance, efficiency, and decision-making.

CORE COMPETENCIES

Simulation & Digital: Rockwell Emulate 3D, Siemens Plant Simulation, Digital Twins, IoT

Data & Systems: SAP, Power BI, Celonis

Engineering: Lean Manufacturing, Process Optimisation, System Integration

PROFESSIONAL EXPERIENCE

Digital Process Automation & Operations Support | Nouryon | Gothenburg *Sept 2025 – Feb 2026*

- Contributed to deployment of an OCR-based PO processing system, reducing manual handling steps and improving order processing consistency across European formats
- Led UAT cycles ensuring system reliability before go-live, acting as a bridge between technical teams and operational end-users to ensure smooth system adoption
- Drove process standardisation by diagnosing data quality and format inconsistencies, enabling consistent data integration into daily operational workflows

Supply Chain & Process Support | Nouryon | Gothenburg *Summer 2025*

- Managed customer order planning aligned with demand forecasts, supporting on-time delivery targets using SAP and Celonis for live process visibility
- Supported automation of order placement process through data validation and gap analysis, reducing manual intervention in routine order cycles

Freelance Data & Process Solutions Developer | Remote | *2024- Present*

- Built a custom Excel-based scheduling tool for a pest control company, eliminating scheduling conflicts and improving planning efficiency
- Developed a Power BI dashboard for KPI visualisation, enabling data-driven operational decisions

Digital Solutions Developer (Salesforce Platform) | Speridian Technologies | *India & Singapore 2021 – 2024*

- Designed a customer service portal that enhanced interaction efficiency and streamlined case resolution workflows
- Delivered on-site client solutions in Singapore, partnering with cross-functional teams to implement CRM customisations — demonstrating ability to translate customer requirements into effective software solutions

SELECTED ACADEMIC PROJECTS

Master's Thesis: Real-Time Human-Centric Digital Twin for Manufacturing 2025 – Present

- Developed a real-time digital twin integrating human motion capture (Occurrence) with Rockwell Emulate3D via MQTT
- Implemented live 3D human proxy replication using 17-joint pose estimation (~20 Hz streaming)
- Designed and implemented a safety monitoring system with zone-based detection and emergency stop logic (<50 ms response)
- Built a resilient communication architecture with MQTT TCP and WebSocket fallback for deployment across different networks

Assembly Workstation Redesign — 18 to 85 Units/Day Output Increase 2024–2025

- Redesigned workstation applying Lean, ergonomics, and Siemens Plant Simulation, increasing daily output from 18 to 85 units within ergonomic and cost constraints.

Optimising Production with Discrete Event Simulation 2024–2025

- Built a DES model in Siemens Plant Simulation to identify top bottlenecks and optimise buffer sizes, resource allocation, and throughput

Compactable Boiler — Additive Manufacturing (DfAM) Jan–Mar 2025

- Designed and scaled prototype from FDM to PBF-LB (SS 316FR) using SolidWorks, Ansys Granta, and Materialise Magics

EDUCATION

Master's in Production Engineering | Chalmers University of Technology |

Gothenburg 2024 – Present

- Key Courses: Manufacturing Processes, Production Systems, Simulation of Production Systems, Production Management. Board Member — Chalmers Entrepreneurship Society.

Bachelor's in Mechanical Engineering | APJ Abdul Kalam Technological University |

India Graduated 2021

- Final Year Project: Crash Guard Modification for Motorbike Safety — FEM analysis and 3D modelling.

TECHNICAL SKILLS

CAD / CAE: SolidWorks, AutoCAD, Ansys

Simulation & Digital Twin: Siemens Plant Simulation, Rockwell Emulate 3D, MATLAB, Simulink, Materialise Magics, Simplify3D

PLM & Digital Thread: Product Lifecycle Management, BOM management, digital thread principles (academic — Chalmers thesis)

Data & Operations: SAP, Celonis, Power BI, Excel (Advanced)

Methodologies: Lean, Six Sigma (White Belt), PDCA, Ergonomics, Value Stream Mapping, FEM, Industry 5.0

CERTIFICATIONS & ACHIEVEMENTS

Certifications: ASNT NDT Level II | QA/QC Mechanical | MATLAB Onramp | Salesforce Admin | Lean Six Sigma White Belt

Achievements: Research Paper — National Conference on Emerging Trends in ME (2021) | Young Innovators Program (YIP 2020), State Level | Patent Application (First Round Cleared) — KSCSTE